

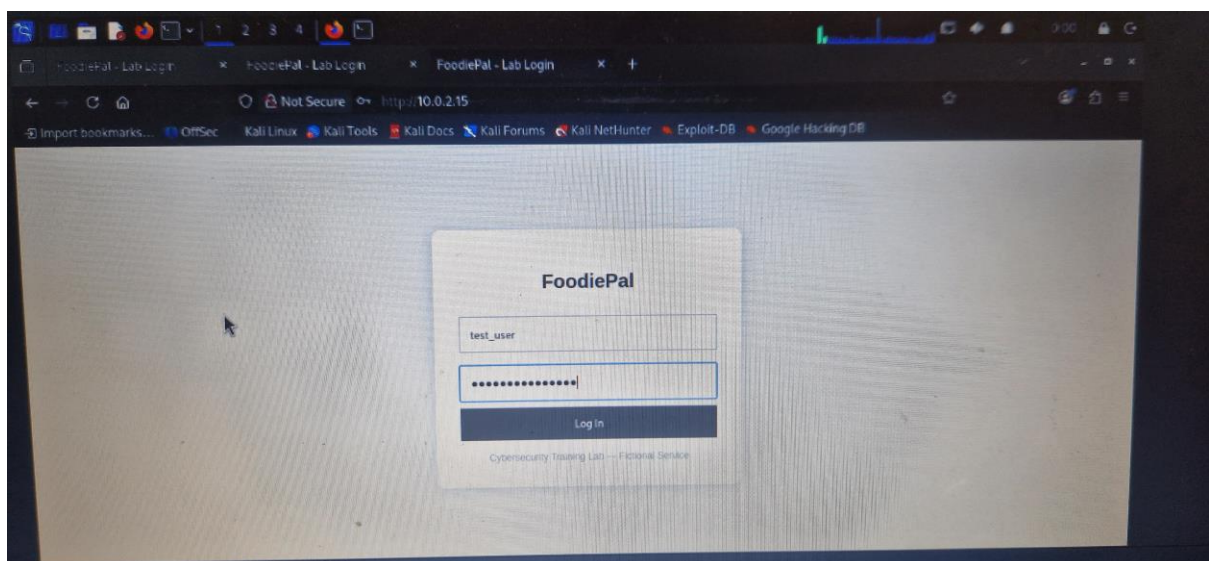
1. Create a table listing 3 real-world examples each of phishing, ransomware, and insider threats that could affect apps like WhatsApp, Flipkart, or Paytm. For each, describe in 1-2 lines how the attack works.

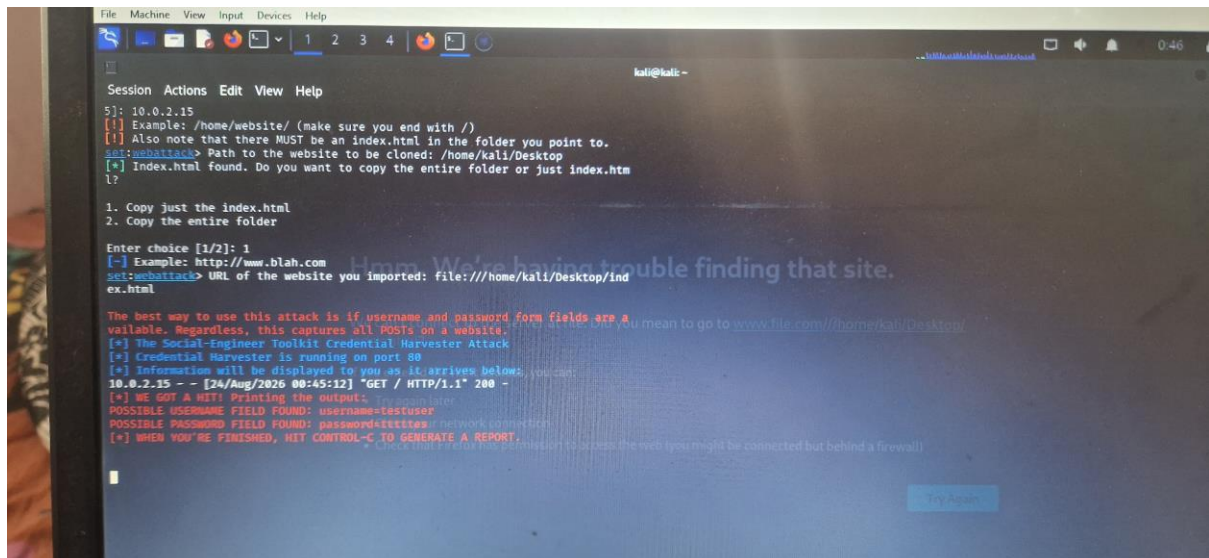
Threat	Real-World example	How the attack works
Phishing	WhatsApp account takeover scams	Attackers send fake WhatsApp messages claiming the user's account needs verification. A link leads to a fake login/verification page that steals credentials or OTPs.
Phishing	Paytm/KYC phishing scams	Victims receive fake SMS/email saying their Paytm KYC or wallet will be blocked. The link directs them to fraudulent page that collects login, card or OTP information.
Phishing	Flipkart fake-delivery/refund scams	Attackers impersonate Flipkart and claim there is a delivery problem or refund waiting. The victim is sent to a fake website or persuaded to provide payment/account details.
Ransomware	Wannacry (2017)	Wannacry exploited a windows vulnerability to spread across networks and encrypt files, demanding cryptocurrency for recovery. A similar attack could disrupt the

		systems supporting an e-commerce or payment service.
Ransomware	Colonial Pipeline (2021)	The ransomware group gained access to the company's network and encrypted systems forcing major operational disruption. An attack on a large online service could similarly take internal systems offline.
Ransomware	Change Healthcare (2024)	Attackers used stolen credentials to gain access and deploy ransomware, causing widespread disruption to healthcare payment and claims processing. This illustrates the risk to organizations handling large volumes of sensitive transactions.
Insider threat	Uber breach (2022)	An attacker obtained an employee's credentials and used them to access internal systems. It demonstrates how compromised or misused employee access can expose sensitive company resources.
Insider threat	Tesla employee data	Former employees were

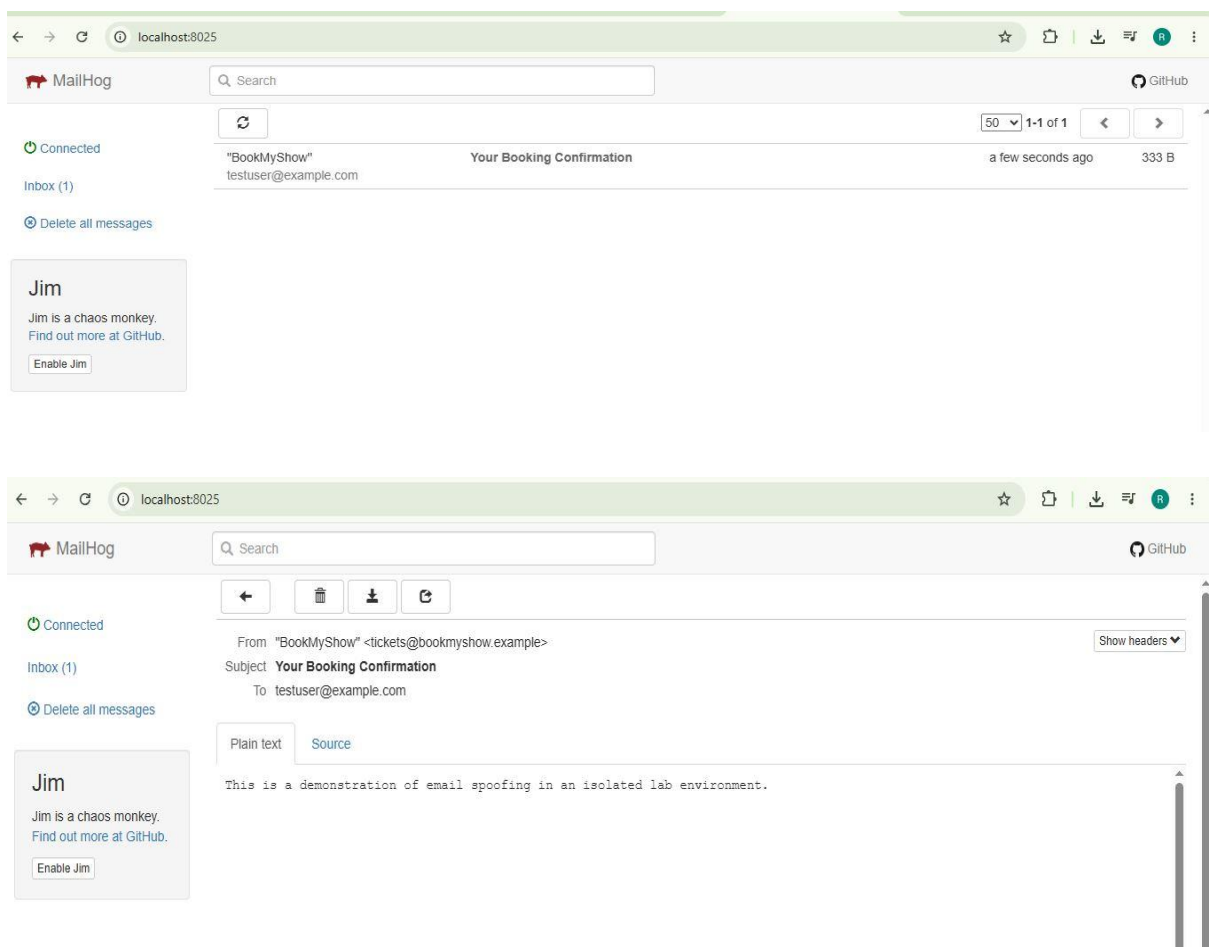
	exposure (2023)	involved in the mishandling of sensitive company information that was later exposed. Excessive or poorly controlled employee access can create significant data-leak risks.
Insider Threat	Morrisons employee data leak (UK)	An employee deliberately disclosed personal information about thousands of customers after obtaining it through their employment. It shows how legitimate internal access can be abused to steal or expose data.

2. Use the Social Engineering Toolkit (SET) to simulate a basic phishing page that mimics a login screen for a popular app (e.g., Instagram or Zomato). Capture and display the entered credentials in your local environment.

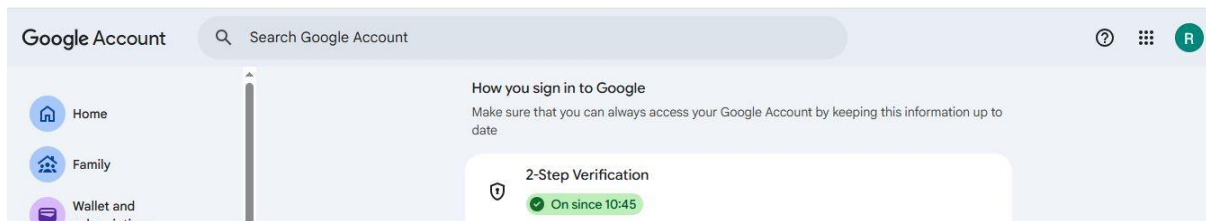




3. Demonstrate email spoofing by sending a test email to yourself using a tool like 'sendmail' or an online SMTP testing service. Show how the sender's name can be faked to look like a message from BookMyShow or IRCTC.



- 4. Research and implement at least two prevention techniques against phishing or email spoofing (such as enabling 2FA or checking email headers) on your own email account. Take screenshots or write a short note explaining how you applied each technique.**



- I enabled 2-step verification on my Gmail account. This adds an additional authentication factor besides my password. Even if a phishing attack tricks me into revealing my password an attacker would still need the second authentication factor to access my account.
- 5. Explain how a phishing attack could target a Zomato or Swiggy user, and write a step-by-step guide for a new user on how to recognize and avoid such attacks. Include at least 3 warning signs to look for in fake emails or websites.**
- A phishing attack against a Zomato or Swiggy user typically tries to trick the user into visiting a fake website or sharing sensitive information such as passwords, OTPs, card details or UPI information.

Example of how the attack works

- 1) The attacker sends a fake SMS or email claiming: “Your Swiggy/Zomato account has been suspended” or “Your refund is waiting”.
- 2) The message contains a link to a fake website that looks similar to the real app/website.
- 3) The user enters their login details or payment information.
- 4) The attacker receives the information and may attempt to misuse the account or payment details.

Step-by-step guide for a new user

Step-1 Check the sender

- Look carefully at the email address or phone number.
- Be suspicious of strange addresses, misspellings or unofficial domains.

Step-2 Check the website address

- Before entering information check the URL in the browser's address bar.
- A fake site might use addresses such as Zomato-security-example.com rather than the company's legitimate domain.
- HTTPS alone does not prove that a website is genuine.

Step-3 Look for urgency

- Be suspicious of messages saying "Act immediately", "Account will be blocked" or "Claim your refund within 10 minutes".
- Attackers use urgency to prevent users from thinking carefully.

Step-4 Never share OTPs or passwords

- Never give an OTP, UPI PIN, password or card PIN to someone who contacts you unexpectedly.
- A legitimate support representative should not ask you to disclose your UPI PIN or OTP.

Step-5 Don't click suspicious links

- Instead of clicking a link in an email or SMS, open the official Zomato/Swiggy app yourself and check for notifications there.

Step-6 Verify unexpected offers or refunds

- If you receive a message about a refund, order cancellation or prize verify it through the official app rather than using the provided link.

Warning Signs

Warning sign	Why it is suspicious
Urgent/threatening language	Tries to make you act without thinking.
Suspicious URL/Domain	Fake websites often imitate the real company's name.
Request for OTP,UPI, PIN or password	Legitimate services should not require you to disclose these secrets.